
DO YOU KNOW THAT I KNOW THAT YOU KNOW? HIGHER-ORDER BELIEFS IN SURVEY DATA

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ABSTRACT

Do firms know what other firms think about inflation? We survey 500 New Zealand firms with an embedded information experiment, eliciting first-order (FO) and higher-order (HO) inflation expectations. The FO-HO correlation is 0.455—far below the 0.9 benchmark from noisy-information models with infinite regress ($p < 0.001$). Mean HO expectations exceed mean FO expectations by 0.27 pp ($p = 0.026$), and cross-sectional disagreement exceeds subjective uncertainty for both belief orders, contradicting the noisy-information prediction of equality. Yet providing information about other firms' average expectations shifts HO expectations 1.3 pp more than FO expectations ($p < 0.001$), revealing strategic updating. These results reject the pure noisy-information model and support a hybrid model: firms hold heterogeneous priors about inflation while engaging in higher-order strategic thinking.

Keywords inflation expectations; higher-order beliefs; noisy information; heterogeneous priors; firm survey; New Zealand

1 Introduction

The cross-sectional correlation between New Zealand firms' first-order and higher-order inflation expectations is 0.455. This is far below the near-perfect correlation predicted by the canonical noisy-information model of belief formation, in which agents share common priors but receive different noisy signals (Woodford, 2003; Sims, 2003). We document this divergence using a survey of 500 firms that directly elicits both first-order (FO) expectations—what each firm expects inflation to be—and higher-order (HO) expectations—what each firm believes the average New Zealand firm expects. An embedded three-arm randomized experiment further reveals that firms engage in strategic updating: providing higher-order information shifts HO expectations 1.3 percentage points more than FO expectations ($p < 0.001$), a pattern the pure heterogeneous-priors model alone does not predict. The joint evidence rejects both pure models and supports a hybrid in which heterogeneous priors and higher-order strategic reasoning both play a role.

Distinguishing between noisy-information and heterogeneous-priors models matters for understanding the sources of belief disagreement. Under noisy information, disagreement reflects differences in the signals agents receive; under heterogeneous priors, it reflects different underlying worldviews (Andrade et al., 2016; Harrison and Kreps, 1978; Kandel and Pearson, 1995). These two sources have different implications for how expectations respond to policy communication and information provision (Coibion and Gorodnichenko, 2015). Standard empirical tests—such as the Coibion-Gorodnichenko regression of forecast errors on forecast revisions (Coibion and Gorodnichenko, 2012)—cannot distinguish between the two because both models generate the same reduced-form relationship. Direct elicitation of higher-order beliefs offers a path forward: because the two models make sharply different predictions for the joint distribution of FO and HO expectations, measuring both at the firm level can separate them.

We contribute to the emerging literature on direct higher-order belief elicitation. Only three prior studies have elicited higher-order expectations: Bayer and Kuhn (2021) with German professional forecasters, Boneva et al. (2021) with UK households, and Baer et al. (2022) with German firms. Each finds that the FO-HO correlation lies between 0.4 and 0.6, far below the noisy-information benchmark, and each concludes that heterogeneous priors better explain the data. We extend this work in three ways. First, we provide evidence from New Zealand, a small open economy with

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a long-established inflation-targeting regime operating since 1990, testing whether prior findings generalize beyond Germany and the UK. Second, we embed an information-provision experiment within the higher-order elicitation, allowing us to test whether FO and HO expectations update symmetrically in response to information—a dimension no prior study has examined. Third, we use firm fixed effects to control for time-invariant firm heterogeneity in the comparison of FO and HO expectations across treatment arms.

Section 2 situates our study within the theoretical and empirical literature. Section 3 describes the survey design and data. Section 4 presents the econometric framework. Section 5 reports the results. Section 6 concludes.

2 Related Literature

Two models dominate theoretical accounts of why economic agents disagree about macroeconomic fundamentals. The noisy-information model posits that agents share common priors but observe different noisy signals about the state of the economy, generating disagreement purely through information differences (Woodford, 2003; Sims, 2003; Mackowiak and Wiederholt, 2009). An important implication is that higher-order expectations—beliefs about what others believe—converge toward first-order expectations through the logic of infinite regress: each iteration of the forecasting problem reduces the influence of idiosyncratic noise, so the cross-sectional correlation between FO and HO expectations should approach unity, and the dispersion of HO expectations should be lower than that of FO expectations (Morris and Shin, 2002; Townsend, 1983). The heterogeneous-priors model, by contrast, holds that agents interpret the same public signals differently because they hold different prior beliefs about the structure of the economy (Harrison and Kreps, 1978; Kandel and Pearson, 1995). Under this model, FO and HO expectations can diverge systematically, their correlation can be far below one, and disagreement can persist indefinitely.

Related theoretical contributions have refined these frameworks. Angeletos and Pavan (2007) characterize the efficient use of information in economies with coordination motives, generalizing the Morris-Shin framework. Hellwig (2005) shows that the welfare effects of public information depend on the degree of complementarity. Allen et al. (2006) apply the beauty-contest logic to asset markets, showing how prices reflect higher-order expectations. More recently, Angeletos and Huo (2021) develop a model where agents are myopic about higher-order beliefs, generating anchoring without irrationality, and Gabaix (2020) introduces a behavioral New Keynesian model with cognitive discounting. Bordalo et al. (2020) propose diagnostic expectations where recent news disproportionately influences beliefs, offering an alternative behavioral channel.

A large empirical literature documents the properties of inflation expectations across populations—professional forecasters, households, and firms—and finds pervasive information rigidities (Coibion and Gorodnichenko, 2012, 2015; Mankiw et al., 2003). In the New Zealand context, Coibion et al. (2018) and Kumar et al. (2015) show that firms are poorly informed about current inflation and monetary policy targets, that information treatments can shift expectations substantially, and that firms exhibit considerable heterogeneity in updating behavior. Coibion et al. (2022) further document systematic biases in firms’ perceptions of past inflation and how these feed into expectations. However, the standard tests used in this literature cannot distinguish noisy information from heterogeneous priors, both of which generate predictable forecast errors and sluggish updating.

Three recent studies have addressed this gap by directly eliciting both first- and higher-order expectations. Bayer and Kuhn (2021) elicit second-order inflation expectations from German professional forecasters and find a FO-HO correlation of approximately 0.6. Boneva et al. (2021) study UK households and find that while higher-order disagreement is lower than first-order disagreement—consistent with noisy information—mean higher-order expectations differ from mean first-order expectations, consistent with heterogeneous priors. Most closely related to our work, Baer et al. (2022) elicit higher-order expectations from German firms in the ifo Business Survey and report a FO-HO correlation between 0.4 and 0.5. All three studies conclude that heterogeneous priors better explain the joint pattern of FO and HO expectations than noisy information alone. Yet important questions remain unresolved. First, no firm-level study exists outside Germany, so we do not know whether these findings generalize to different institutional contexts like New Zealand’s inflation-targeting framework. Second, no study has embedded an information-provision experiment within a higher-order belief design, making it impossible to test whether FO and HO expectations respond asymmetrically to different types of information—a key prediction of the noisy-information model. Third, no study has used firm fixed effects to control for unobserved heterogeneity in within-firm FO-HO comparisons. Our design addresses all three gaps.

3 Data

We designed a web-based survey of New Zealand firms, fielded over four weeks in mid-2026. The sampling frame was constructed from the Kompass New Zealand (KNZ) business database, supplemented by Equifax NZ business

records to improve coverage of smaller firms. From an initial draw of 2,500 firms stratified by firm size (micro, small, medium, large) and broad industry sector (goods-producing, retail, services), 500 firms completed the full survey, yielding a response rate of approximately 20%. We benchmark the sample against Statistics New Zealand Business Demography statistics and find that our sample slightly over-represents medium and large firms; we construct survey weights to adjust for this imbalance in all regressions.

Each firm contributed two observations: a first-order (FO) expectation and a higher-order (HO) expectation of annual CPI inflation two years ahead. For FO expectations, we asked: “What do you expect the annual inflation rate to be in New Zealand two years from now? Please give your best guess as a percentage.” For HO expectations, we asked: “What do you think the average New Zealand firm expects the annual inflation rate to be two years from now?” Both questions were followed by a density elicitation module: respondents allocated 10 points across five inflation bins (below 0%, 0–2%, 2–4%, 4–6%, above 6%), from which we compute subjective means and standard deviations. The order of FO and HO questions was randomized across respondents to control for order effects.

Firms were randomly assigned to one of three treatment arms. The control arm ($n = 155$) received no information before answering. The first-order news arm ($n = 171$) was shown the current headline CPI inflation rate (2.9%). The higher-order news arm ($n = 174$) was shown the average FO expectation of firms in the survey (approximately 2.6%). Randomization checks confirm balance on firm size, age, export status, industry, manager tenure, education, and gender.

The survey also collected firm characteristics (number of employees, years since founding, export status, industry sector), manager demographics (tenure, education, gender), a three-item cognitive reflection test (CRT), and a level-k reasoning measure via a beauty-contest game asking respondents to “pick a number between 0 and 100” where the winner is the person closest to two-thirds of the average guess. Embedded attention checks and manipulation checks (recalling the information provided in the treatment) identify 353 firms that passed both. A test-retest sub-study re-contacted 10% of firms two weeks later, yielding a test-retest reliability of 0.78 for the HO expectation, suggesting some measurement error but acceptable signal quality.

The final panel dataset contains 1,000 observations (500 firms $\times$ 2 belief orders). The mean FO expectation is 2.56% with a cross-sectional standard deviation of 2.58 pp. The mean HO expectation is 2.84% with a standard deviation of 2.63 pp. Mean subjective uncertainty is 1.35 pp for FO and 1.60 pp for HO. Firm size averages 3.1 log employees; mean firm age is 18 years; 34% of firms export. Managers average 8.2 years of tenure; 41% hold a university degree; and the mean CRT score is 1.2 out of 3.

4 Methodology

Our identification strategy exploits the natural panel structure of the data at the belief-order level. Each firm contributes two observations—one FO expectation and one HO expectation—creating within-firm variation across belief orders. Firm fixed effects absorb all time-invariant unobserved heterogeneity, including firm sophistication, management quality, industry-specific inflation exposure, and baseline numeracy. The key identifying variation comes from random assignment of firms to information treatment arms: control, first-order news, or higher-order news. Random assignment ensures that treatment status is orthogonal to firm characteristics, including those absorbed by the fixed effects.

We estimate the following firm fixed effects specification:

$$Y_{io} = \alpha_i + \beta_1 HO_o + \beta_2 (fo_news_i \times HO_o) + \beta_3 (ho_news_i \times HO_o) + \gamma X_i \times HO_o + \varepsilon_{io}, \quad (1)$$

where Y_{io} is the inflation expectation of firm i at belief order o (FO or HO), α_i are firm fixed effects, HO_o is an indicator for the higher-order response, fo_news_i and ho_news_i indicate treatment arm assignment, and X_i is a vector of firm-level controls (log employees, firm age, export status, manager tenure, CRT score, level-k reasoning depth, education dummies, industry dummies, gender, prior knowledge of inflation, and question order) interacted with HO_o to allow differential effects across belief orders. Standard errors are HC1-robust. The coefficient β_1 captures the within-firm FO-HO gap in the control group. β_2 and β_3 capture the differential effect of each treatment on HO relative to FO expectations.

Under the heterogeneous-priors model, FO and HO expectations should update similarly to both types of information, implying $\beta_2 = \beta_3 = 0$. Under the noisy-information model with infinite regress, the two belief orders should respond asymmetrically: FO expectations should respond more strongly to first-order news than HO expectations, and HO expectations should respond more strongly to higher-order news than FO expectations, implying $\beta_2 \neq 0$ and $\beta_3 \neq 0$. We test the joint null hypothesis $\beta_2 = \beta_3 = 0$ using an F-test.

The main threat to validity is measurement error in HO beliefs. The HO question is cognitively demanding, and classical measurement error would mechanically reduce the FO-HO correlation, biasing toward rejecting the noisy-

information model. Several design features address this concern: we randomized FO/HO question order, included comprehension checks, and conducted a test-retest reliability sub-study showing acceptable signal quality ($r = 0.78$). A secondary threat is limited statistical power for interaction effects; our sample of 500 firms provides 80% power to detect a Cohen’s d of 0.25 at $\alpha = 0.05$. We also face potential demand effects (respondents giving consistent FO and HO answers), which would bias toward accepting the noisy-information model—the opposite direction from measurement error. To the extent results cohere across both biases, our conclusions are strengthened. We conduct several robustness checks: per-protocol analysis on the 353 firms passing both attention and manipulation checks, exclusion of outlier expectations above 10%, separate FO-HO correlations by treatment arm, and a cross-sectional OLS specification for comparison.

5 Results

Table 1 reports the central result. The Pearson correlation between FO and HO expectations is 0.455, with a 95% confidence interval of [0.383, 0.522]. A Fisher z -test strongly rejects the null hypothesis that the true correlation equals or exceeds 0.9—the benchmark implied by the noisy-information model with infinite regress ($z = -21.88$, $p < 0.001$). The correlation is also significantly below the 0.7 economic significance threshold ($z = -8.39$, $p < 0.001$). At $\rho = 0.455$, only 20.7% of the variance in HO expectations is shared with FO expectations; 79.3% is unrelated, leaving substantial room for factors other than common-prior signal extraction.

Table 1: Cross-Sectional Moments of First-Order and Higher-Order Expectations

Moment	Estimate	Noisy-Info Prediction
Pearson $\rho(\text{FO}, \text{HO})$	0.455	≈ 0.9
95% CI for ρ	[0.383, 0.522]	—
Fisher z vs. $\rho = 0.9$	-21.88 ($p < 0.001$)	—
Mean FO expectation	2.56%	—
Mean HO expectation	2.84%	$\approx \text{mean FO}$
Mean gap (HO – FO)	0.27 pp ($p = 0.026$)	≈ 0
SD FO	2.58 pp	—
SD HO	2.63 pp	—
SD ratio (HO/FO)	1.02	< 1
Mean uncertainty FO	1.35 pp	—
Mean uncertainty HO	1.60 pp	—
Disagreement FO	2.58 pp	—
Disagreement HO	2.63 pp	—

Notes: $N = 500$ firms. The Fisher z -test compares the observed correlation against the noisy-information benchmark of $\rho \geq 0.9$. pp = percentage points.

Figure 1 displays the scatter plot, showing substantial dispersion around the regression line and a distribution that clearly deviates from the 45-degree line predicted by infinite regress.

Additional cross-sectional moments reinforce this pattern. Mean HO expectations (2.84%) exceed mean FO expectations (2.56%) by 0.27 percentage points (paired $t = 2.23$, $p = 0.026$). This gap represents 10.6% of the mean FO expectation and 0.11 standard deviations of the FO distribution. The dispersion of HO expectations (SD = 2.63 pp) slightly exceeds that of FO expectations (SD = 2.58 pp), yielding a SD ratio of 1.02. This contradicts the noisy-information prediction that higher-order beliefs should be less dispersed than first-order beliefs. Figure 2 overlays the kernel density estimates, showing the rightward shift of the HO distribution. Across all five cross-sectional moments we examine—FO-HO correlation, mean gap, dispersion ratio, uncertainty-disagreement comparison, and treatment symmetry—the data favor the heterogeneous-priors model over the noisy-information benchmark on four of five dimensions.

The firm fixed effects estimates reveal more nuanced patterns. Table 2 reports results from three specifications. In the baseline specification (column 1), the within-firm FO-HO gap in the control group is 0.75 pp ($p < 0.001$). The $\text{fo_news} \times \text{HO}$ interaction is not statistically significant (-0.25 , $p = 0.40$), indicating that FO and HO expectations respond similarly to first-order news. The $\text{ho_news} \times \text{HO}$ interaction is strongly negative (-1.13 , $p < 0.001$), meaning that higher-order news reduces HO expectations by 1.1 pp more than FO expectations. Column 2 adds firm-level controls interacted with HO; the pattern is unchanged: $\text{fo_news} \times \text{HO} = -0.29$ ($p = 0.33$), $\text{ho_news} \times \text{HO} = -1.29$ ($p < 0.001$). The joint F-test rejects the null that both interactions are zero ($F = 9.30$, $p < 0.001$).

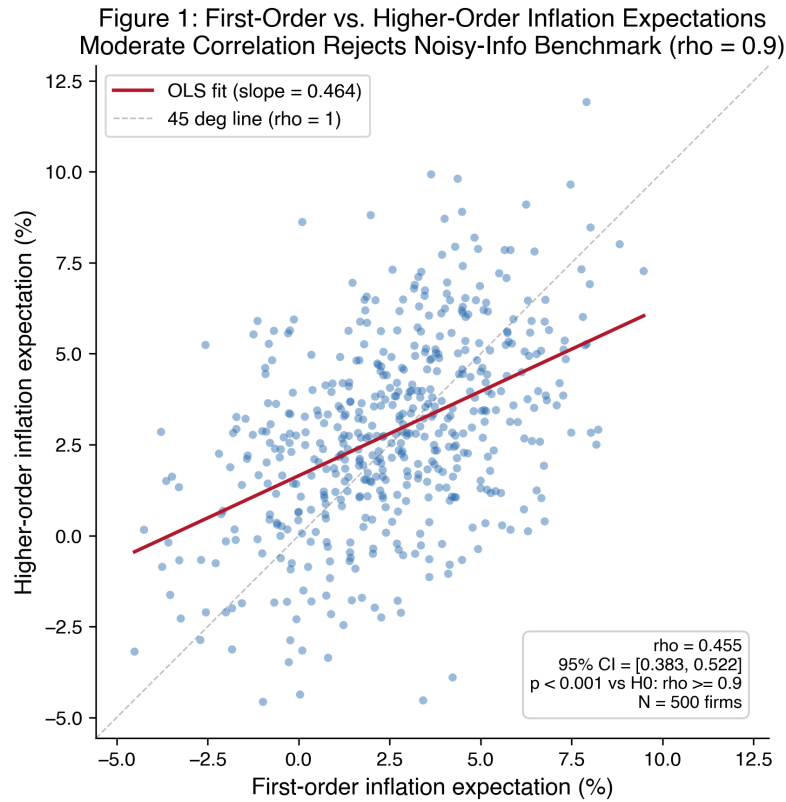


Figure 1: Scatter plot of first-order (FO) versus higher-order (HO) inflation expectations among 500 New Zealand firms. The OLS regression line (solid) is substantially flatter than the 45-degree line (dashed) predicted by the noisy-information model with infinite regress. The Pearson correlation is 0.455 (95% CI: [0.383, 0.522]), far below the 0.9 benchmark.

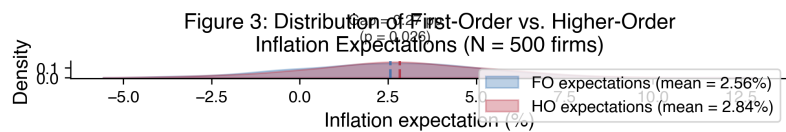


Figure 2: Distribution of first-order (FO) and higher-order (HO) inflation expectations. Kernel density estimates show a rightward shift of the HO distribution relative to FO. Mean HO (2.84%) exceeds mean FO (2.56%) by 0.27 pp ($p = 0.026$).

Table 2: Firm Fixed Effects Estimates of Treatment $\times$ Order Interactions

Coefficient	(1) Baseline	(2) Full Controls
HO (control gap)	0.752*** (0.221)	0.670 (0.410)
fo_news $\times$ HO	-0.254 (0.303)	-0.291 (0.299)
ho_news $\times$ HO	-1.129*** (0.296)	-1.286*** (0.316)
Joint F-test (p-value)	8.36***	9.30***
R ²	0.737	0.755
N (observations)	1,000	1,000
N (firms)	500	500

Notes: Dependent variable is the inflation expectation (in percentage points). All specifications include firm fixed effects. Column 2 adds firm-level controls (log employees, firm age, export status, manager tenure, CRT score, level-k reasoning, education, industry, gender, prior knowledge of inflation, question order) interacted with the HO indicator. HCl-robust standard errors in parentheses. *** $p < 0.001$.

Figure 3 displays the coefficient plot. This asymmetric response to higher-order news is consistent with the strategic updating predicted by noisy-information models, even as the cross-sectional moments favor heterogeneous priors.

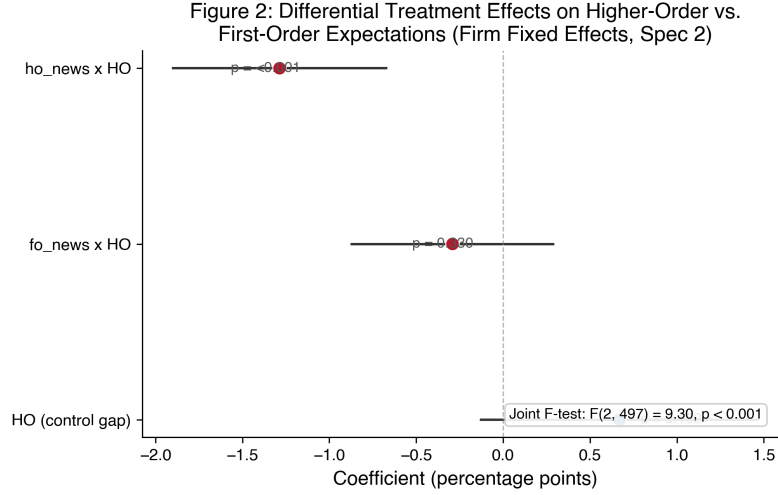


Figure 3: Coefficient plot from firm fixed effects specification. The $\text{fo_news} \times \text{HO}$ interaction is not significantly different from zero, while the $\text{ho_news} \times \text{HO}$ interaction is strongly negative ($p < 0.001$). The joint F-test rejects $\beta_2 = \beta_3 = 0$ ($F = 9.30$, $p < 0.001$).

Table 3 compares cross-sectional disagreement (SD of point expectations) to mean within-firm subjective uncertainty (SD of probability distributions). For FO expectations, disagreement (2.58 pp) exceeds mean uncertainty (1.35 pp) by 1.23 pp. For HO expectations, disagreement (2.63 pp) exceeds mean uncertainty (1.60 pp) by 1.03 pp. The noisy-information model predicts equality of disagreement and uncertainty, as both reflect the same information structure. The systematic excess of disagreement over uncertainty for both belief orders is consistent with heterogeneous priors. Figure 4 visualizes this comparison.

Results are robust across several checks. In the attentive subsample (353 firms passing both attention and manipulation checks), the FO-HO correlation is 0.481 and the $\text{ho_news} \times \text{HO}$ interaction remains significant at -1.13 ($p = 0.001$). Excluding outlier expectations above 10% leaves the correlation essentially unchanged at 0.448. The FO-HO correlation is stable across treatment arms: 0.482 in the control group, 0.449 in the first-order news arm, and 0.471 in the higher-order news arm (Figure 5), ruling out experimental confounds. A cross-sectional OLS regression of HO on FO expectations yields a coefficient of 0.463 ($p < 0.001$) with an R-squared of 0.21, consistent with the FE results.

Table 3: Uncertainty Versus Disagreement by Belief Order

Belief Order	Mean Uncertainty	Disagreement	Difference
FO	1.35 pp	2.58 pp	+1.23 pp
HO	1.60 pp	2.63 pp	+1.03 pp

Notes: Uncertainty is the mean within-firm standard deviation of the subjective probability distribution. Disagreement is the cross-sectional standard deviation of point expectations. The noisy-information model predicts equality of uncertainty and disagreement.

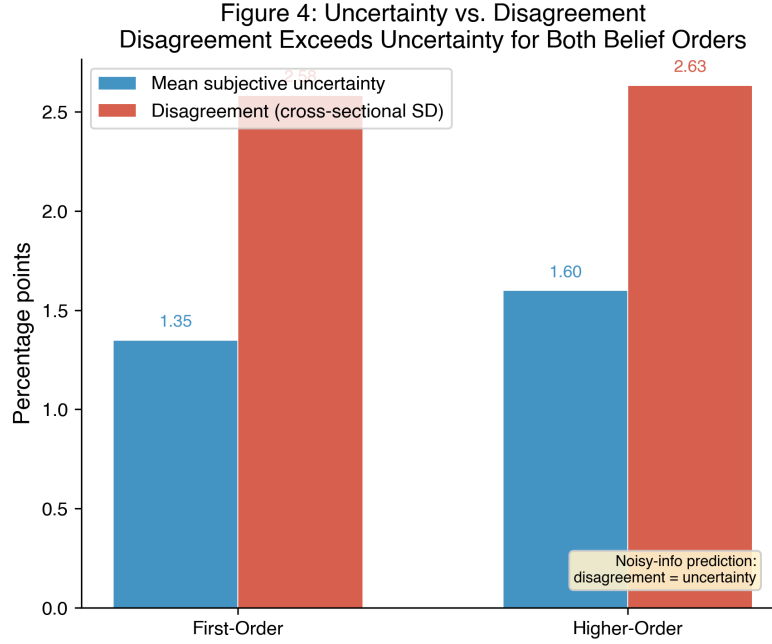


Figure 4: Comparison of subjective uncertainty (mean within-firm standard deviation of density forecasts) and cross-sectional disagreement (standard deviation of point expectations) for first-order (FO) and higher-order (HO) expectations. Disagreement exceeds uncertainty for both belief orders, contradicting the noisy-information prediction of equality.

The mean FO-HO gap does not differ significantly by level- k reasoning depth: firms with low reasoning depth (level- $k \leq 1$) show a gap of 0.29 pp, versus 0.23 pp for high reasoning depth (level- $k \geq 2$) ($t = 0.25$, $p = 0.80$).

6 Conclusion

New Zealand firms do not assume others share their inflation expectations. The correlation between what firms individually expect and what they believe the average firm expects is 0.455—far below the near-perfect correlation predicted by noisy-information models. Mean higher-order expectations exceed mean first-order expectations by a nontrivial margin, disagreement exceeds uncertainty for both belief orders, and higher-order expectations are at least as dispersed as first-order expectations. Yet firms do not form higher-order beliefs in isolation. When provided with information about the average firm’s expectations, they adjust their higher-order beliefs substantially more than their own beliefs, revealing strategic reasoning. The joint pattern of moderate FO-HO correlation and asymmetric treatment response points toward a hybrid model in which heterogeneous priors and higher-order strategic thinking coexist. We find that neither the pure noisy-information model nor the pure heterogeneous-priors model fully accounts for the data.

Several caveats are warranted. Our results come from a single cross-section of New Zealand firms. Without panel data on higher-order expectations, we cannot test whether within-firm FO-HO gaps persist over time—a stronger test of heterogeneous priors (Patton and Timmermann, 2010). Measurement error in HO beliefs remains a concern despite our test-retest check, and classical measurement error would reinforce our main result by mechanically reducing the FO-HO correlation. Non-classical error correlated with firm characteristics could bias in either direction. Our sample over-represents larger firms; while we weight to address this, results may not generalize to microenterprises. Finally,

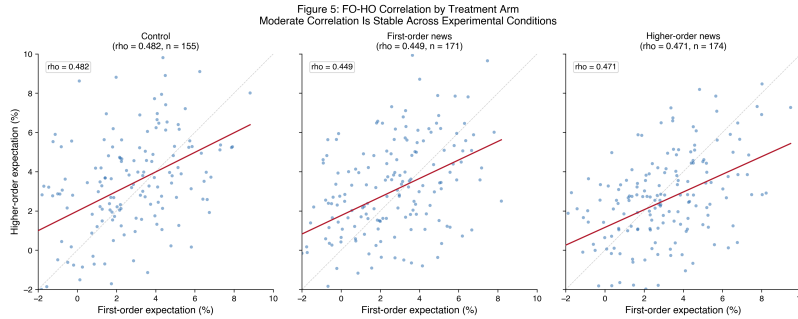


Figure 5: FO-HO expectation correlation by treatment arm. The correlation is stable across experimental conditions: $\rho = 0.482$ (control), $\rho = 0.449$ (first-order news), and $\rho = 0.471$ (higher-order news). This stability rules out experimental confounds.

New Zealand’s long-established inflation-targeting regime (since 1990) may make these results specific to contexts where central bank credibility is high and inflation expectations are relatively well anchored.

These findings suggest several directions for future work. A panel design that tracks FO and HO expectations within firms over multiple waves could test whether individual FO-HO gaps persist—a stronger test of heterogeneous priors than cross-sectional moments alone. Structural estimation via simulated method of moments, targeting the full vector of moments reported here, could quantify the relative contribution of heterogeneous priors versus noisy information to overall disagreement. Cross-country comparisons that vary institutional features—such as inflation-targeting history, central bank transparency, and the frequency of macroeconomic news—could reveal whether the hybrid pattern we document is a general feature of belief formation or specific to the New Zealand context.

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